



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-2027

Course Handout (Part-II)

Date: 3/08/2026

In addition to Part-I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : PHAF413

Course Title : Pharmaceutical Management and Quality Control

Instructor-in-charge : Prof. Murali M Pandey

1. Scope & Objective of the Course: It is a course intended to give the students an idea of various aspects of pharmaceutical management with emphasis on latest trends in current good manufacturing practices and international certification. This course also deals with process validation, quality control, pilot plant and scale up operations, drug regulatory affairs, production management, pharmaceutical marketing, management of materials and finance and management information systems.

2. Text Book:

T1: O.P. Khanna, Industrial Engineering and Management, Dhanpat Rai Publication, Revised Edition, 1999

T2: Willing, H.Sidney, Tuckerman, M. Murray, Hitchings IV, S. William: Good Manufacturing practices for Pharmaceuticals, Marcel Dekker, 4th ed.

3. Ref. Book:

R1: Subba Rao Chaganti, Pharmaceutical Marketing in India, Pharma Book Syndicate, 2007.

R2: Lachmann, Liebermann & Kanig: the Theory & Practice of Industrial, Pharmacy, K.M. Varghese, Third Edition, 1990. New Delhi.

R3: International Organisation for Standardization Geneva: 1580 9000 Quality Management Systems 2nd Ed. Pub. By International Trade Center Geneva.

R4: Bernard. T. Loftus, Robert A. Nash: Pharmaceutical Process Validation, Marcel Dekker.



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4. Course Plan :

Module	Lecture session/Tutorial Session.	Reference	Learning Outcome
GMP guidelines	Introduction	T2, chapter 3	Understanding of good manufacturing practices regulations for pharmaceutical industries.
	Design & features of cGMP unit	T2, chapter 4	
	Equipment design & features	T2, chapter 5	
	Air control systems	T2, chapter 5	
	Process control requirements for cGMP unit	T2, chapter 7	
	Record maintenance	T2, chapter 11	
	Packaging & labeling requirements	T2, chapter 8	
Validation	Validation-basics & methodology	R4, chapter 1	Understanding of validation of process and equipment to meet the regulatory requirements.
	Validation of solid dosage forms	R4, chapter 5,6	
	Prospective validation	R4, chapter 7	
	Retrospective validation	R4, chapter 8	
	Sterile product validation	R4, chapter 4	
Quality control tools	Quality Control & Assurance	R2, chapter 27	Role of QC and QA in Pharmaceuticals
ISO	Introduction and Salient features of ISO	R2, chapter 1,2	ISO certification understanding
	Interpretation of various clauses of ISO	R3, Class notes	
Quality by Design	QbD aspects and guidelines in pharmaceutical product development	Class notes	Understanding of QbD in product development.
Pilot Plant and scale up	Pilot plant requirements & scale up operations	T1,chapter 23	Understanding about production scale up
Statistical applications in pharmaceuticals	Statistical analysis for decision making	R2, chapter 10	Application of statistics in decision
Pharmaceutical Management	1. Introduction and importance of Drug Discovery, Product Development, and Pharmaceutical Management.	T.1,Chapter 2; Class notes	Understanding the pharmaceutical management at industry from
	Production management	T1, chapter 1,15	



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	Inventory Control and Management	T1,chapter 24	product proposal to product life cycle.
	Budgeting	T1, chapter 28	
	Pharmaceutical Marketing Management	R1, chapter 3	
	Product Costing, Break Even analysis	T1, chapter 27	
	Market Research	R.1, chapter 3 Class notes	

5. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Remarks
Mid-semester test	90 min	25		CB
Continues assessment (Quizzes /case studies / assignments / presentation)	-	30	Throughout the semester	CB/OB
Compre Exam	180 min	45		CB/OB

6. Mid-Semester Evaluation: Will be announced after the Mid--semester test.

7. Make-up Policy:

The facility of make-up test is meant to take care of ‘Unavoidable’ absence from scheduled tests. It is expected that the students will “keep faith” in this respect and avoid any misuse of this useful feature. There will be no insistence on “certificate” but all decisions to give or not to give make-up tests would be based on mutual faith and trust.

8. Chamber Consultation Hour: To be announced in the class.

9. Notices: Notices, if any, concerning the course will be informed through email.

Instructor In-Charge
PHA F413



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